

1. Consider the series $\sum_{n=2}^{\infty} ne^n$.
 - a) Explain how we can know that this series diverges?
 - b) If this series is changed to $\sum_{n=2}^{\infty} ne^{-n}$ it now converges. Find a way to prove this series converges.
2. Use any of the techniques discussed in class thus far to determine if the following series converge or diverge. If possible, determine what the series converges to.
 - a) $\sum_{k=1}^{\infty} \frac{3^{k+1}}{6^{2k}}$
 - b) $\sum_{k=1}^{\infty} \frac{k+1}{k^{3/2} + \sqrt{k}}$
 - c) $\sum_{k=2}^{\infty} \frac{e^k}{\ln(k)}$
 - d) $\sum_{k=1}^{\infty} \left(\frac{3^{k+1}}{5^k} + \frac{k^2}{k^4} \right)$
 - e) $\sum_{k=1}^{\infty} \sin^2(k) \cos(k)$
 - f) $\sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1} \right)$
3. Consider the series $\sum_{n=0}^{\infty} \frac{7n}{n^2 + 5}$. Suppose we wish to apply the Integral Test to this series.
 - a) Show conditions for the Integral Test work from the 3rd term onward.
 - b) Show behavior of series from 3rd term onward.
 - c) Make conclusion about full series.
4. Find all values of c for which the following series converges $\sum_{n=1}^{\infty} \left(\frac{c}{n} - \frac{1}{n+1} \right)$.